

CHE

210

Hw #5

Q5.

$V_0 = 1000 \text{ L}$ of 95 wt% ethanol
5 wt% water mix @ 20°C
0.5 m³ initially

$m_{\text{in}} = m_{\text{out}} = 100 \text{ L/min}$ of 95 wt% ethanol
inflow = pure water @ $t=0$ & $Q_{\text{in}} = 100 \text{ L/min}$

$$\rho_w = 0.99823 \text{ kg/L}$$

$$\rho_{95} = 0.80424$$

$$\rho_5 = 0.98938 \text{ kg/L}$$

$$\dot{m}_1 = \dot{m}_0$$

$$\dot{m}_1 = \rho_w Q_1 = (0.99823 \text{ kg/L}) \left(\frac{100 \text{ L}}{1 \text{ min}} \right) = 99.823 \frac{\text{kg}}{\text{min}} = \dot{m}_0$$

SUB

$$w = \frac{m_{\text{eth.}}}{M}$$

$$\frac{dm_{\text{ethanol}}}{dt} = 0 - w \dot{m}_0 = -w \dot{m}_0$$

$$\frac{dm_{\text{eth.}}}{dt} = -\frac{\dot{m}_0}{M} m_{\text{eth.}} \Rightarrow \frac{1}{m_{\text{eth.}}} \times dm_{\text{eth.}} = -\frac{\dot{m}_0}{M} dt \Rightarrow m_{\text{eth.}}(t) = m_{\text{eth.}0} e^{\left(-\frac{\dot{m}_0}{M}\right)t}$$

✓

$$w(t) = w_0 e^{\left(-\frac{\dot{m}_0}{M}\right)t}$$

$$V_0 = 1000 \text{ L}$$

$$M = \rho_{95} V_0 = (0.80424 \text{ kg/L}) \left(\frac{1000 \text{ L}}{1} \right) = 804.24 \text{ kg}$$

Need $w(t) = 0.05$ from $w_0 = 0.95$

$$\ln \left(\frac{0.05}{0.95} \right) = \frac{0.95}{0.95} e^{\left(\frac{99.823}{804.24} t \right)}$$

$$t = \frac{804.24}{99.823} \ln \left(\frac{0.95}{0.05} \right)$$

$$t = (8.056) \ln(19) = (8.056) (2.944) = 23.7 \text{ min}$$

$$y_0 = w_0 \rho_{95} = (0.95)(0.80424) = 0.764026 \text{ kg/L}$$

NEED

$$y_t = w_t \rho_5 = (0.05)(0.98938) = 0.049469 \frac{\text{kg}}{\text{L}}$$

$$\frac{Q}{V} = \frac{100}{1000} = 0.1 \text{ min}^{-1}$$

$$y_t = y_0 e^{-0.1t} \Rightarrow t = \frac{1}{0.1} \ln \left(\frac{y_0}{y_t} \right)$$

$$t = 10 \ln \left(\frac{0.764026}{0.049469} \right) = 27.4 \text{ min}$$

✓

30/30

$$w(t) = w_0 e^{\left(-\frac{\dot{m}_0}{M}\right)t}$$

$$V_0 = 1000 \text{ L}$$

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$$Q_{\text{out}} = Q_{\text{in}} = 100 \text{ L/min}$$

$$V = 1000 \text{ L} \quad \text{mass} \neq \text{conserved}$$

$$\dot{m}_{\text{eth.},0} = w \dot{m}_0 = w(\rho(w)Q)$$

Ethanol Balance:

$$\frac{dm_{\text{eth.}}}{dt} = -w \rho(w) Q$$

$$M_{\text{eth.}} = w M = w(\rho(w)V)$$

$$\dot{m}_{\text{eth.}} = V \left(w \frac{d\rho(w)}{dt} \right)$$

$$\text{DIFF.} \Rightarrow \frac{d m_{\text{eth.}}}{dt} = V \frac{d}{dt} (w \rho(w))$$

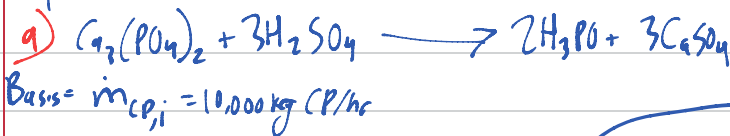
$$V \frac{d}{dt} (w \rho(w)) = -w \rho(w) Q$$

$$\frac{dy}{dt} = -Q y$$

$$\frac{dy}{y} = \left(\frac{-Q}{1} \right) dt$$

$$y(t) = y_0 e^{-Q t} \quad y(t) = w \rho(w)$$

Q4.



$\text{mm} \text{ Ca}_3(\text{PO}_4)_2 = 316.17 \frac{\text{g}}{\text{mol}}$
 $\text{H}_2\text{SO}_4 = 98.06 \frac{\text{g}}{\text{mol}}$
 $\text{H}_3\text{PO}_4 = 97.99 \frac{\text{g}}{\text{mol}}$
 $\text{CaSO}_4 = 136.14 \frac{\text{g}}{\text{mol}}$
 $\text{H}_2\text{O} = 18.015 \frac{\text{g}}{\text{mol}}$

$\dot{n}_{\text{SA}} = 3 \dot{n}_{\text{CP}} = 3(37.24) = 96.72 \frac{\text{kmol}}{\text{hr}}$

$\dot{m}_{\text{SA}, \text{pure}} = 96.72 \frac{\text{kmol}}{\text{hr}} \left(\frac{98.06 \text{ kg}}{\text{kmol}} \right) = 9486 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_{\text{H}_2\text{SO}_4} = 9486 \text{ kg/hr}$

95 wt % $\rightarrow \dot{m}_2 = \frac{\dot{m}_{\text{H}_2\text{SO}_4}}{0.95} = \frac{9486}{0.95} = 9984 \frac{\text{kg}}{\text{hr}}$ ✓

Water in stream 2 $\rightarrow \dot{m}_{\text{W},2} = 0.05 \dot{m}_2 = 0.05(9984) = 499 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_{\text{PA}} = 2 \dot{m}_{\text{CP}} = 2(37.24) = 64.48 \text{ kmol/hr}$

$\dot{m}_{\text{PA}} = 64.48 \frac{\text{kmol}}{\text{hr}} \left(\frac{97.99 \text{ kg}}{\text{kmol}} \right) = 6319 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_3 = \frac{\dot{m}_{\text{PA}}}{0.4} = \frac{6319}{0.4} = 15797 \frac{\text{kg}}{\text{hr}}$ ✓

$\dot{m}_{\text{W},3} = 0.6 \dot{m}_3 = 0.6(15797) = 9478 \frac{\text{kg}}{\text{hr}}$

$\dot{n}_{\text{CS}} = 3 \dot{n}_{\text{CP}} = 96.72 \text{ kmol/hr}$

$\dot{m}_{\text{CS}} = 96.72 \frac{\text{kmol}}{\text{hr}} \left(\frac{136.14 \text{ kg}}{\text{kmol}} \right) = 13167 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_{\text{W},4} = 2 \dot{m}_{\text{CS}} = 2(96.72) = 193.44 \frac{\text{kmol}}{\text{hr}}$

$\dot{m}_{\text{W},4} = 193.44 \frac{\text{kmol}}{\text{hr}} \cdot \frac{18.015 \text{ kg}}{\text{kmol}} = 3485 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_4 = \dot{m}_{\text{CS}} + \dot{m}_{\text{W},4} = 13167 + 3485 = 16652 \frac{\text{kg}}{\text{hr}}$ ✓

$\text{CP} = 10,000 \frac{\text{kg}}{\text{hr}}$

Water = $\dot{m}_{\text{W},1}$

$\dot{m}_{\text{W},1} + \dot{m}_{\text{W},2} = \dot{m}_{\text{W},3} + \dot{m}_{\text{W},4}$

$\dot{m}_{\text{W},1} + 499 = 9478 + 3485$

$\dot{m}_{\text{W},1} = 12464 \frac{\text{kg}}{\text{hr}} \rightarrow \dot{m}_1 = 10,000 + 12,464 = 22,464 \frac{\text{kg}}{\text{hr}}$ ✓

$\dot{m}_{\text{in}} = \dot{m}_1 + \dot{m}_2 = 22464 + 9985 = 32,449 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_0 = \dot{m}_3 + \dot{m}_4 = 15797 + 16652 = 32,449 \frac{\text{kg}}{\text{hr}}$

b) $M = 10,000 \text{ kg}$

$X(t) = \text{mass fraction}$

Initial: $X(0) = 0.2$

$\dot{m}_{\text{PA}, \text{gen}} = 6319 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_3 = 15797 \frac{\text{kg}}{\text{hr}}$

$\text{PA}_i = 0, \text{PA}_0 = X(t) \dot{m}_3, \text{PA}_{\text{gen}} = \dot{m}_{\text{PA}, \text{gen}}$

$M \frac{dx}{dt} = \dot{m}_{\text{PA}, \text{gen}} - \dot{m}_3 X$

$\frac{dx}{dt} + \left(\frac{\dot{m}_3}{M} \right) X = \frac{\dot{m}_{\text{PA}, \text{gen}}}{M}$

$\frac{dx}{dt} = 0 = \dot{m}_{\text{PA}, \text{gen}} - \dot{m}_3 X_{ss} \Rightarrow X_{ss} = \frac{\dot{m}_{\text{PA}, \text{gen}}}{\dot{m}_3}$
 @ S.S

$X_{ss} = \frac{6319}{15797} = 0.40$

$X(t) = X_{ss} + (X_0 - X_{ss}) e^{\left(\frac{-\dot{m}_3}{M} t \right)}$

$t = 1 \text{ hr} \quad X(1) = 0.40 + (0.2 - 0.4) e^{\frac{-15797}{10000} \cdot 1} \approx 0.206$ ✓

0.359 mass fraction PA
 0.35.9 wt %

29/29

Q1. 3.21

$$\frac{1}{n} \cdot \frac{dn}{dt} = -0.116 n \cdot dt$$

$$\int \frac{1}{n} dn = \int -0.116 dt$$

$$\ln(n) = -0.116t + C$$

$$n(t) = e^{(-0.116t + C)}$$

$$e^C = n_0 \Rightarrow n_0 = n(t)$$

$$\Delta m(t) = m_0 e^{-0.116t}$$

$$m_0 = 200 \text{ mg}$$

$$m(t) = 200 e^{-0.116t} \text{ mg}$$

$$n(t) = 100 \text{ mg}$$

$$\frac{100}{200} = \frac{200 e^{-0.116t}}{200}$$

$$0.5 = e^{-0.116t}$$

$$\ln(0.5) = -0.116t$$

$$t = \frac{-\ln(0.5)}{0.116} = \frac{\ln(2)}{0.116} \quad h = 5.975 \checkmark$$

$t=0$ @ 6am $\rightarrow$

$t=8h \Rightarrow 2\text{pm}$

24 hr interval $\Rightarrow 0 \leq t \leq 24$

$$m(t) = 200 e^{-0.116t} \text{ mg} \quad (0 \leq t \leq 8)$$

$\hookrightarrow$ 6am - bar not day

@ $t=8$ (2pm)

Before $m(8) = 200 e^{-0.116 \cdot 8}$

After $m(8) = 200 e^{-0.116 \cdot 8} + 200$

$$m(t) = 200 e^{-0.116t} + 200 e^{-0.116(t-8)} \text{ mg} \quad (8 \leq t \leq 24)$$

$\hookrightarrow$ 11pm $\Rightarrow$
 $t=17h$

$$m(17) = 200 e^{-0.116 \cdot 17} + 200 e^{-0.116(17-8)} = 200 e^{-1.972} + 200 e^{-1.048} =$$

$$\rightarrow = 98.24438 \dots$$

skip 2pm mug

$$m(17) = 200 e^{-0.116 \cdot 17} = 200 e^{-1.972} = 27.8356$$

$$\Delta m = 98.2443 - 27.8356 = 70.4087 \text{ mg}$$

Yes, cutting the 2nd mug reduces
11pm caffeine load by abt 70 mg

No plot? -5

15
20

3.29

Q2

$N(t)$ = # of immunotoxin molecules in cancer cell @ time t (mol.)
 $N(0) = 0$

enters cell = 62,000 mol/h

leaves = 57,000 mol/h

degrades: $2700(1 - e^{-0.3t})$ mol/h

$$\frac{dN}{dt} = 62000 - 57000 - 2700(1 - e^{-0.3t})$$

$$62000 - 57000 = 5000 \text{ mol/h}$$

$$\frac{dN}{dt} = 5000 - 2700 + 2700e^{-0.3t} = 2300 + 2700e^{-0.3t} \quad N(0) = 0$$

$$N(t) - N(0) = \int_0^t (2300 + 2700e^{-0.3s}) ds$$

$$N(t) = \underbrace{\int_0^t 2300 ds}_{2300t} + \underbrace{\int_0^t 2700e^{-0.3s} ds}_{2700 \left[\frac{1 - e^{-0.3t}}{0.3} \right]}$$

$$N(t) = 2300t + 9000(1 - e^{-0.3t})$$

$$N(8) = 2300(8) + 9000(1 - e^{-0.3(8)}) = 26,583.538 \text{ molecules}$$

Threshold:
30,000 mol.

NO, you shouldn't invest
bc $N(8) \neq$ reach
30,000 mol.

30,000 mol
3416.465 mol
after 8 hrs
accumulation
= below
required amt

15/15

Q3

No problem 3, -20